Finite Pauli–Villars Products and the Eta Invariant

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Abstract

We give an explicit spectral construction of Pauli–Villars regularization for geometric Dirac operators on closed odd-dimensional spin manifolds. A finite collection of regulator masses, with binomial multiplicities, defines an ordinary absolutely convergent product at every positive regulator scale. Using the Bismut–Freed local heat-kernel estimate, we prove that its phase converges to $\exp(-\mathrm{i}\pi\eta_D(0)/2)$. For this particular scheme, the modulus also has a finite limit, equal to the positive square root of the zeta determinant of D^2 when D is invertible. An antiunitary symmetry with square -1 gives a compatible Pfaffian construction and the phase $\exp(-\mathrm{i}\pi\eta_D(0)/4)$. We also treat negative physical mass and zero modes, and isolate the analytic hypotheses needed beyond the odd-dimensional oriented setting. The purpose is to provide a direct convergence proof of the established eta-phase formula, with the regulator scheme and the Pfaffian branch specified explicitly.

1 Introduction

The phase of a fermion partition function contains information that is not determined by its absolute value. For a massless complex fermion, a familiar Pauli–Villars (PV) prescription leads to the expression

$$Z = |Z| \exp\left(-\frac{\mathrm{i}\pi}{2}\eta_D(0)\right), \tag{1.1}$$

where D is a self-adjoint Dirac operator and the sign depends on the regulator-mass convention. This formula, and its Pfaffian counterpart, play a central role in the discussion of fermion anomalies and topological phases in [1, 2].

To make a statement such as (1.1) precise, one must specify both the regularized determinant and the limiting operation. For example, the formal product $\prod_\lambda \lambda/(\lambda + \mathrm{i}M)$ is not an ordinary absolutely convergent product in three dimensions. Taking a large-mass limit eigenvalue by eigenvalue does not justify interchanging that limit with an infinite spectral sum. Moreover, a Pfaffian requires a compatible choice of sign, rather than an unspecified square root of a determinant.

We address these points by an explicit finite PV construction. For any even integer N larger than the dimension, the coefficients

$$c_j = (-1)^j \binom{N}{j}, \qquad 0 \le j \le N, \tag{1.2}$$

cancel all moments of degree less than N. Grouping the regulator factors at each eigenvalue then produces an ordinary Fredholm determinant. Its phase admits a convergent heat-kernel representation. The Bismut–Freed estimate for the local eta density supplies an integrable majorant, so that the large-mass limit follows from dominated convergence. The same moment conditions also cancel the divergent terms in the modulus in odd dimensions.

The analytic input is the local regularity theorem of Bismut and Freed [3]; an alternative proof is given by Ponge [4]. The geometry of determinant and Pfaffian lines, including families and boundary problems, is developed in [5, 6]. The present paper concerns explicit scalar spectral products for a fixed operator and does not replace that geometric framework. The phase formula itself is established in the literature. Our aim is to exhibit a concrete finite-regulator realization and to justify each interchange of a sum, an integral, and a large-mass limit.

Section 2 records the geometric and analytic assumptions. Section 3 states and proves the determinant theorem. Section 4 gives the Pfaffian construction. Negative mass and zero modes are treated in Section 5. Section 6 discusses extensions and limitations.

2 Geometric setting and analytic input

Let Y be a closed Riemannian spin manifold of odd dimension $n = 2r + 1$, and let $E \to Y$ be a Hermitian vector bundle with a unitary connection. Write $S \to Y$ for the complex spinor bundle. We consider the geometric twisted Dirac operator

$$D : H^1(Y; S \otimes E) \longrightarrow L^2(Y; S \otimes E)$$

as a self-adjoint unbounded operator on $L^2(Y; S \otimes E)$. Here "geometric" refers to the operator formed from the Levi–Civita and twisting connections, without an additional arbitrary zero-order potential. The regulator convention throughout is

$$D + \mathrm{i}m, \qquad m > 0. \tag{2.1}$$

Replacing every regulator mass by its negative conjugates the massless phase obtained below.

Spectral sums count eigenvalues with their complex multiplicities. In particular, the independent Grassmann variables of a complex fermion and its conjugate do not introduce a second copy of this spectral multiplicity. When a kernel is present, zero eigenvalues are omitted from the eta function:

$$\eta_D(s) = \sum_{\lambda \neq 0} \operatorname{sgn}(\lambda) |\lambda|^{-s}, \qquad \Re s > n. \tag{2.2}$$

We set $h = \dim_{\mathbb{C}} \ker D$ and use $\operatorname{ph} z = z/|z|$ for $z \neq 0$.

Introduce the heat traces

$$K(t) = \operatorname{Tr}(e^{-tD^2}), \qquad T(t) = \operatorname{Tr}(De^{-tD^2}), \qquad A(t) = \operatorname{Tr}(|D|e^{-tD^2}). \tag{2.3}$$

The Weyl estimate implies

$$\#\{\lambda : 0 < |\lambda| \leq R\} = O(R^n), \qquad A(t) = O(t^{-(n+1)/2}) \quad (t \downarrow 0). \tag{2.4}$$

Both $T(t)$ and $A(t)$ decay exponentially as $t \to \infty$ because the nonzero spectrum has a positive gap from zero. The same is true of $K(t)$ when $h = 0$.

Theorem 2.1 (Bismut–Freed local regularity)**.** *For the geometric Dirac operator above, the fiberwise trace of the kernel of De^{-tD^2} on the diagonal is uniformly $O(t^{1/2})$ as $t \downarrow 0$. Consequently,*

$$T(t) = O(t^{1/2}), \qquad \int_0^\infty t^{-1/2} |T(t)| \, \mathrm{d}t < \infty. \tag{2.5}$$

This is the analytic result of [3, Theorem 2.4]; see also [4, Theorem 3.1]. We use it as an established theorem.

Lemma 2.2. *The eta function is holomorphic for $\Re s > -2$ and satisfies*

$$\eta_D(0) = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} T(t)\, \mathrm{d}t, \tag{2.6}$$

where the integral is absolutely convergent.

Proof. For $\Re s > n$, absolute convergence permits the Mellin representation

$$\eta_D(s) = \frac{1}{\Gamma((s+1)/2)} \int_0^\infty t^{(s-1)/2} T(t)\, \mathrm{d}t. \tag{2.7}$$

Theorem 2.1 and exponential decay at infinity show that the integral is locally uniformly convergent for $\Re s > -2$. It therefore gives a holomorphic continuation to that half-plane. Evaluation at $s = 0$ gives (2.6). □

The argument below needs the absolute integrability in (2.5), not merely regularity of an abstract eta function at $s = 0$. This distinction matters when considering more general self-adjoint elliptic operators.

3 The finite PV determinant

Assume first that $h = 0$. Choose an even integer $N > n$ and take the coefficients (1.2). For $M > 0$ and $\lambda \neq 0$, set

$$g_M(\lambda) = \prod_{j=0}^{N} (\lambda + \mathrm{i}jM)^{c_j}, \qquad Z_M(D) = \prod_{\lambda \in \operatorname{spec} D} g_M(\lambda). \tag{3.1}$$

The factor with $j = 0$ is the physical massless determinant. Positive and negative coefficients specify positive and inverse determinant powers, respectively. The factors are combined at each eigenvalue before the infinite product is taken.

Write the short-time expansion of the heat trace as

$$K(t) = \sum_{k=0}^{r} a_{2k} t^{k-n/2} + O(t^{1/2}), \qquad Q(t) = K(t) - \sum_{k=0}^{r} a_{2k} t^{k-n/2}. \tag{3.2}$$

Define

$$\mathcal{A}(D) = \exp\left[-\frac{1}{2} \int_0^\infty Q(t) \frac{\mathrm{d}t}{t} \right]. \tag{3.3}$$

The integral converges at both endpoints: $Q(t)/t = O(t^{-1/2})$ near zero, while $K(t)$ decays exponentially and the subtracted powers decay at infinity.

Theorem 3.1 (Determinant limit)**.** *For every $M > 0$, the product (3.1) is absolutely convergent and nonzero. Moreover, $g_M(D) - 1$ is trace class and*

$$Z_M(D) = \det{}_F\, g_M(D). \tag{3.4}$$

Its large-mass limit is

$$\lim_{M \to \infty} Z_M(D) = \mathcal{A}(D) \exp\left(-\frac{\mathrm{i}\pi}{2} \eta_D(0) \right). \tag{3.5}$$

In particular,

$$\lim_{M \to \infty} \operatorname{ph} Z_M(D) = \exp\left(-\frac{\mathrm{i}\pi}{2} \eta_D(0) \right). \tag{3.6}$$

Both limits are independent of the even integer $N > n$ within this family of regulator schemes.

Corollary 3.2 (Three-dimensional formula)**.** *If $n = 3$, one may take $N = 4$ and obtain*

$$Z_M(D) = \prod_{\lambda \in \operatorname{spec} D} \frac{\lambda(\lambda + 2\mathrm{i}M)^6(\lambda + 4\mathrm{i}M)}{(\lambda + \mathrm{i}M)^4(\lambda + 3\mathrm{i}M)^4}. \tag{3.7}$$

This product satisfies (3.5) *with*

$$\mathcal{A}(D) = \exp\left[-\frac{1}{2}\int_0^\infty \frac{K(t) - a_0 t^{-3/2} - a_2 t^{-1/2}}{t}\,\mathrm{d}t\right]. \tag{3.8}$$

The regulator data in this example are

j	0	1	2	3	4
c_j	1	-4	6	-4	1
m_j	0	M	$2M$	$3M$	$4M$

Only the combined product is used. We do not define the individual divergent determinants and then assume multiplicativity of their regularized values.

3.1 Absolute convergence at finite mass

The finite-difference identity gives

$$\sum_{j=0}^{N} c_j p(j) = 0 \quad \text{for every polynomial } p \text{ of degree less than } N. \tag{3.9}$$

In particular, $\sum c_j = 0$ and $\sum c_j j^k = 0$ for $1 \le k < N$. Fix the logarithm

$$\ell_M(\lambda) = \sum_{j=1}^{N} c_j \operatorname{Log}\left(1 + \frac{\mathrm{i}jM}{\lambda}\right), \tag{3.10}$$

where every Log is the principal logarithm. Its argument has real part one, so no factor meets the branch cut. Since $\sum c_j = 0$, $e^{\ell_M(\lambda)} = g_M(\lambda)$ for either sign of λ.

For fixed M, expansion at large $|\lambda|$ and (3.9) give

$$\ell_M(\lambda) = O_M(|\lambda|^{-N}). \tag{3.11}$$

The Weyl estimate and $N > n$ imply

$$\sum_{\lambda} |\ell_M(\lambda)| < \infty. \tag{3.12}$$

It follows that the unordered product is well defined and nonzero, with

$$Z_M(D) = \exp\left(\sum_{\lambda} \ell_M(\lambda)\right). \tag{3.13}$$

Also $\sum_\lambda |g_M(\lambda) - 1| < \infty$. In a spectral basis this is precisely the trace-class condition for $g_M(D) - 1$, and it proves (3.4).

3.2 A convergent formula for the phase

The logarithm (3.10) fixes a real-valued lift of the phase:

$$\Theta_M = \operatorname{Im} \sum_{\lambda} \ell_M(\lambda) = \sum_{\lambda} \sum_{j=1}^{N} c_j \arctan \frac{jM}{\lambda}. \tag{3.14}$$

The arctangent takes values in $(-\pi/2, \pi/2)$. For $a \in \mathbb{R}$ and $\lambda \neq 0$,

$$\begin{aligned} \arctan \frac{a}{\lambda} &= \int_0^a \frac{\lambda}{\lambda^2 + u^2} \, \mathrm{d}u \\ &= \int_0^\infty \lambda e^{-t\lambda^2} \left(\int_0^a e^{-tu^2} \, \mathrm{d}u \right) \mathrm{d}t. \end{aligned} \tag{3.15}$$

The inner integral is oriented when $a < 0$. Introduce

$$H_M(t) = \sum_{j=1}^{N} c_j \int_0^{jM} e^{-tu^2} \, \mathrm{d}u. \tag{3.16}$$

At fixed M, its expansion at zero is

$$H_M(t) = \sum_{q \geq 0} \frac{(-t)^q M^{2q+1}}{q!(2q+1)} \sum_{j=1}^{N} c_j j^{2q+1}. \tag{3.17}$$

Since N is even, the terms with $0 \leq q < N/2$ vanish. Hence

$$H_M(t) = O_M(t^{N/2}). \tag{3.18}$$

Together with (2.4), this gives

$$\int_0^\infty \sum_{\lambda} |H_M(t) \lambda e^{-t\lambda^2}| \, \mathrm{d}t = \int_0^\infty |H_M(t)| A(t) \, \mathrm{d}t < \infty. \tag{3.19}$$

Indeed, the integrand is $O_M(t^{(N-n-1)/2})$ at zero and is integrable at infinity. Thus Fubini's theorem applies after the PV terms have been grouped in H_M, and yields

$$\Theta_M = \int_0^\infty H_M(t) T(t) \, \mathrm{d}t. \tag{3.20}$$

For fixed $t > 0$, $\sum_{j=1}^{N} c_j = -1$ implies

$$H_M(t) \longrightarrow -\frac{\sqrt{\pi}}{2\sqrt{t}} \quad (M \to \infty). \tag{3.21}$$

Uniformly in M,

$$|H_M(t)| \leq (2^N - 1) \frac{\sqrt{\pi}}{2\sqrt{t}}. \tag{3.22}$$

The integrability in (2.5) therefore permits dominated convergence in (3.20). By Lemma 2.2,

$$\lim_{M \to \infty} \Theta_M = -\frac{\sqrt{\pi}}{2} \int_0^\infty t^{-1/2} T(t) \, \mathrm{d}t = -\frac{\pi}{2} \eta_D(0). \tag{3.23}$$

This proves the phase statement. In fact, the chosen real lift itself converges; the conclusion is stronger than convergence modulo 2π. Heat kernels have only been used to analyze the already defined PV product, rather than as an additional prescription for defining it.

3.3 The modulus and cancellation of local terms

Set

$$B_M(t) = \sum_{j=0}^{N} c_j e^{-j^2 M^2 t}. \tag{3.24}$$

The even moments in (3.9) give $B_M(t) = O_M(t^{N/2})$ at zero. For each nonzero eigenvalue, the Frullani identity and $\sum c_j = 0$ imply

$$\sum_{j=0}^{N} c_j \log(\lambda^2 + j^2 M^2) = -\int_0^\infty e^{-t\lambda^2} B_M(t) \frac{\mathrm{d}t}{t}. \tag{3.25}$$

The summed absolute integrand is integrable because

$$|B_M(t)|K(t)/t = O_M(t^{(N-n)/2-1}) \quad (t \downarrow 0),$$

and $K(t)$ decays exponentially at infinity. Consequently,

$$\log|Z_M(D)| = -\frac{1}{2}\int_0^\infty B_M(t)K(t)\frac{\mathrm{d}t}{t}. \tag{3.26}$$

Lemma 3.3 (Vanishing heat moments)**.** *For every odd integer p with $1 \le p \le n$,*

$$\int_0^\infty t^{-1-p/2} B_M(t)\,\mathrm{d}t = \Gamma(-p/2)M^p \sum_{j=1}^{N} c_j j^p = 0. \tag{3.27}$$

All integrals in the left-hand expression are ordinary convergent integrals of the combined function B_M.

Proof. Convergence follows from $N > p$ at zero and boundedness of B_M at infinity. Let $q = (p+1)/2$. Integrating by parts q times gives

$$\frac{(-1)^q}{\prod_{\ell=0}^{q-1}(-p/2+\ell)} \int_0^\infty t^{q-1-p/2} B_M^{(q)}(t)\,\mathrm{d}t. \tag{3.28}$$

At each step the boundary terms vanish: at zero one uses the vanishing order of B_M and its derivatives; at infinity the initial power is negative and all subsequent derivatives decay exponentially. After differentiating, the $j = 0$ term is absent and each remaining exponential is individually integrable. Evaluating these integrals and using the recurrence of the gamma function gives the middle expression in (3.27). The odd moment in (3.9) then makes it zero. □

Each subtracted heat term in (3.2) corresponds to $p = n - 2k$, a positive odd integer. Lemma 3.3 thus turns (3.26) into the exact identity

$$\log|Z_M(D)| = -\frac{1}{2}\int_0^\infty B_M(t)Q(t)\frac{\mathrm{d}t}{t}. \tag{3.29}$$

Since $|B_M(t)| \le 2^N$, $B_M(t) \to 1$ for every $t > 0$, and $Q(t)/t$ is absolutely integrable, dominated convergence yields

$$\lim_{M\to\infty} |Z_M(D)| = \mathcal{A}(D) > 0. \tag{3.30}$$

Combining this with (3.23) proves Theorem 3.1 and Corollary 3.2.

In three dimensions, the two cancellations read

$$\int_0^\infty t^{-3/2} B_M(t)\,\mathrm{d}t = -2\sqrt{\pi} M \sum_{j=1}^{4} c_j j = 0, \tag{3.31}$$

$$\int_0^\infty t^{-5/2} B_M(t)\,\mathrm{d}t = \frac{4\sqrt{\pi}}{3} M^3 \sum_{j=1}^{4} c_j j^3 = 0. \tag{3.32}$$

Thus the subtractions in the proof are exact zero identities at finite M. No divergent real counterterm has been added to this particular determinant prescription.

3.4 Identification of the limiting modulus

For comparison with a conventional normalization, define

$$\zeta_{D^2}(s) = \sum_{\lambda} (\lambda^2)^{-s}, \qquad \det{}_\zeta D^2 = \exp[-\zeta'_{D^2}(0)].$$

Proposition 3.4. *For invertible D in the setting of Theorem 3.1,*

$$\mathcal{A}(D) = (\det{}_\zeta D^2)^{1/2}. \tag{3.33}$$

Proof. Put $p_k = n - 2k$ for $0 \le k \le r$. The meromorphic continuation of the Mellin transform is, near $s = 0$,

$$\begin{aligned}\Gamma(s)\zeta_{D^2}(s) &= \int_0^1 t^{s-1} Q(t)\,\mathrm{d}t + \int_1^\infty t^{s-1} K(t)\,\mathrm{d}t + \sum_{k=0}^{r} \frac{a_{2k}}{s - p_k/2} \\ &= \int_0^\infty t^{s-1} Q(t)\,\mathrm{d}t. \end{aligned} \tag{3.34}$$

The second equality follows by integrating the subtracted powers over $[1, \infty)$. The last integral is holomorphic in the strip $-1/2 < \Re s < 1/2$. Since $1/\Gamma(s) = s + O(s^2)$,

$$\zeta'_{D^2}(0) = \int_0^\infty Q(t) \frac{\mathrm{d}t}{t}.$$

Equation (3.33) follows from (3.3). □

This comparison is a consequence of the convergent construction; zeta regularization was not used to define the finite PV product.

4 Pfaffians and Majorana fermions

A Pfaffian formula requires additional geometric structure. We give a sufficient condition that specifies both the antisymmetric form and a compatible square root of the determinant.

Theorem 4.1 (Pfaffian limit)**.** *Assume the hypotheses of Theorem 3.1. Suppose that a fiberwise antilinear isometry induces an antiunitary operator J on the Hilbert space, preserves the domain of D, and satisfies*

$$J^2 = -1, \qquad JD = DJ. \tag{4.1}$$

Take the Hermitian inner product to be linear in its second argument and define

$$\omega(u, v) = \langle Ju, v\rangle, \qquad A_m(u, v) = \omega(u, (D + \mathrm{i}m)v). \tag{4.2}$$

Then A_m is a skew-symmetric complex bilinear form. The finite spectral Pfaffian ratios with coefficients c_j have an absolutely convergent limit $P_M(D)$ as the spectral cutoff is removed, with

$$P_M(D)^2 = Z_M(D). \tag{4.3}$$

Furthermore,

$$\lim_{M\to\infty} P_M(D) = \mathcal{A}(D)^{1/2} \exp\left(-\frac{\mathrm{i}\pi}{4}\eta_D(0)\right). \tag{4.4}$$

The eta invariant here counts the full complex multiplicity, including each pair enforced by J.

Proof. Antiunitarity and $J^2 = -1$ imply that ω is nondegenerate, complex bilinear, and skew-symmetric. Self-adjointness and $JD = DJ$ give

$$\omega(Du, v) = \langle JDu, v\rangle = \langle DJu, v\rangle = \langle Ju, Dv\rangle = \omega(u, Dv). \tag{4.5}$$

Hence A_m is also skew-symmetric.

Let V_λ be the eigenspace associated with a distinct real eigenvalue λ. It is preserved by J and has even complex dimension, say $\dim_{\mathbb{C}} V_\lambda = 2r_\lambda$. The restriction of ω to V_λ is nondegenerate, and

$$\mathrm{Pf}(A_m|_{V_\lambda}) = (\lambda + \mathrm{i}m)^{r_\lambda}\,\mathrm{Pf}(\omega|_{V_\lambda}). \tag{4.6}$$

Use the same basis for every mass on this eigenspace. In the product with exponents c_j, the factor $\mathrm{Pf}(\omega|_{V_\lambda})$ cancels because $\sum c_j = 0$. Basis-change factors cancel for the same reason. Thus the regularized Pfaffian is

$$P_M(D) = \prod_{\lambda \text{ distinct}} g_M(\lambda)^{r_\lambda} = \exp\left(\frac{1}{2} \sum_{\lambda \text{ with multiplicity}} \ell_M(\lambda)\right). \tag{4.7}$$

Absolute convergence follows from (3.12). This proves (4.3) and fixes its square-root branch through finite-dimensional Pfaffian ratios. Its lifted phase is $\Theta_M/2$. Equations (3.23) and (3.30) prove (4.4). □

For example, the usual spin Dirac operator in dimension $8k + 3$, twisted by a bundle with a compatible real structure, has the quaternionic symmetry relevant to this construction; see [6, Section 1.3]. The existence of a Majorana field with the required scalar mass term depends on the dimension and gauge representation. Condition (4.1) is sufficient for the theorem and is not a classification of all real fermion theories.

There are two useful checks on the construction. First, when a single J-pair of eigenvalues crosses zero from negative to positive, the discrete jump in $\eta_D(0)$ is four. The phase in (4.4) changes sign, as a Pfaffian should. A pointwise principal square root of the complex determinant would not in general retain this information.

Second, one cannot obtain an inverse Pfaffian by putting a skew-symmetric quadratic form on a single set of commuting variables. In our scheme every negative coefficient is even. Indeed, $c_j < 0$ requires j odd, while

$$j\binom{N}{j} = N\binom{N-1}{j-1}$$

shows that $\binom{N}{j}$ is even when N is even and j is odd. The inverse Pfaffians can therefore be paired into inverse determinants. The spectral definition itself is already complete without a separate formal functional-integral representation.

5 Negative physical mass and zero modes

5.1 The negative-mass phase

The large negative-mass phase has twice the exponent of the massless formula. This can be proved by a shifted version of the same finite-difference scheme. We now allow $h = \dim_{\mathbb{C}} \ker D$ to be nonzero, retaining the convention (2.2) for eta.

Theorem 5.1 (Negative physical mass)**.** *Let D be as in Section 2, let $N > n$ be even, and take c_j as in (1.2). Define*

$$a_j = 2j - 1, \qquad W_M(D) = \prod_{\lambda \in \operatorname{spec} D} \prod_{j=0}^{N} (\lambda + \mathrm{i} a_j M)^{c_j}. \tag{5.1}$$

Thus the physical mass is $-M$ and all regulator masses are positive. For every $M > 0$, this is an absolutely convergent nonzero product, including its factors at zero eigenvalues, and

$$\lim_{M \to \infty} \operatorname{ph} W_M(D) = \exp[-\mathrm{i}\pi(\eta_D(0) + h)]. \tag{5.2}$$

If a symmetry J satisfying (4.1) exists, the corresponding Pfaffian ratio obeys

$$\lim_{M \to \infty} \operatorname{ph} W_M^{\mathrm{Maj}}(D) = \exp\left[-\frac{\mathrm{i}\pi}{2}(\eta_D(0) + h)\right]. \tag{5.3}$$

Proof. Since a_j is affine in j, $\sum c_j a_j^k = 0$ for $0 \le k < N$. For $\lambda \neq 0$, the logarithm

$$\sum_{j=0}^{N} c_j \operatorname{Log}(1 + \mathrm{i} a_j M/\lambda)$$

is $O_M(|\lambda|^{-N})$. This proves convergence of the nonzero spectral product. Its phase is handled by replacing H_M with

$$\widetilde{H}_M(t) = \sum_{j=0}^{N} c_j \int_0^{a_j M} e^{-tu^2} \, \mathrm{d}u. \tag{5.4}$$

The same moment cancellations give $\widetilde{H}_M(t) = O_M(t^{N/2})$ at zero, while

$$|\widetilde{H}_M(t)| \le 2^N \frac{\sqrt{\pi}}{2\sqrt{t}}. \tag{5.5}$$

As $a_0 = -1$ and $a_j > 0$ for $j \ge 1$,

$$\lim_{M \to \infty} \widetilde{H}_M(t) = \frac{\sqrt{\pi}}{2\sqrt{t}} \sum_{j=0}^{N} c_j \operatorname{sgn}(a_j) = -\frac{\sqrt{\pi}}{\sqrt{t}}. \tag{5.6}$$

Fubini's theorem and dominated convergence apply exactly as before. The nonzero modes contribute $\exp(-\mathrm{i}\pi\eta_D(0))$ for the determinant and $\exp(-\mathrm{i}\pi\eta_D(0)/2)$ for the Pfaffian.

A zero eigenvalue contributes the finite, strictly negative factor

$$\prod_{j=0}^{N} (\mathrm{i} a_j M)^{c_j} = -\prod_{j=1}^{N} a_j^{c_j}. \tag{5.7}$$

The determinant therefore receives an additional sign $(-1)^h$. In the Pfaffian case h is even, and the corresponding sign is $(-1)^{h/2}$. These signs give (5.2) and (5.3). □

In terms of the reduced eta invariant $\bar\eta_D = (\eta_D(0) + h)/2$, these phases are $e^{-2\pi i\bar\eta_D}$ and $e^{-\pi i\bar\eta_D}$, respectively. Here $\bar\eta_D$ denotes the specified real number; in the Pfaffian expression, reducing it modulo integers would discard a sign. These statements concern the phase. The modulus of this massive product generally requires a normalization by real local terms. The cancellation proved for the massless modulus in Section 3 does not assert a corresponding massive limit.

5.2 Massless determinants with the kernel omitted

If $h > 0$, the massless determinant itself vanishes and has no phase. One may nevertheless define $Z'_M(D)$ by the product (3.1) over nonzero eigenvalues only. All phase arguments remain valid, since zero modes do not contribute to $T(t)$.

Proposition 5.2. *For the massless primed product,*

$$\lim_{M\to\infty} \operatorname{ph} Z'_M(D) = \exp\!\left(-\frac{i\pi}{2}\eta_D(0)\right). \tag{5.8}$$

There is a finite positive constant $C_N(D)$ such that

$$M^{-h}|Z'_M(D)| \longrightarrow C_N(D). \tag{5.9}$$

Under (4.1), the primed Pfaffian has phase limit $\exp(-i\pi\eta_D(0)/4)$ *and* $M^{-h/2}|P'_M(D)| \to C_N(D)^{1/2}$.

Proof. The phase follows from (3.20) because the heat trace $T(t)$ is unchanged by removal of the kernel. For the modulus, let

$$S(t) = \sum_{k=0}^{r} a_{2k} t^{k-n/2}, \qquad R(t) = K(t) - h - S(t) + h\mathbf{1}_{(0,1)}(t).$$

Then $R(t)/t$ is absolutely integrable: at zero $R(t) = O(t^{1/2})$, and at infinity $K(t) - h$ decays exponentially while $S(t)$ decays as negative powers. The same finite-mass calculation and Lemma 3.3 give

$$\log|Z'_M(D)| = -\frac{1}{2}\int_0^\infty B_M(t)R(t)\frac{dt}{t} + \frac{h}{2}\int_0^1 B_M(t)\frac{dt}{t}. \tag{5.10}$$

The first integral has a finite limit by dominated convergence. Write $b(u) = \sum_{j=0}^{N} c_j e^{-j^2 u}$, so that $B_M(t) = b(M^2 t)$. Since $b(u) = O(u^{N/2})$ at zero and $b(u) - 1$ decays exponentially at infinity,

$$\int_0^1 B_M(t)\frac{dt}{t} - 2\log M \longrightarrow \beta_N := \int_0^1 \frac{b(u)}{u}\,du + \int_1^\infty \frac{b(u)-1}{u}\,du. \tag{5.11}$$

Thus (5.9) holds with

$$C_N(D) = \exp\!\left[-\frac{1}{2}\int_0^\infty R(t)\frac{dt}{t} + \frac{h}{2}\beta_N\right]. \tag{5.12}$$

The Pfaffian assertion follows by the eigenspace construction of Theorem 4.1. A fixed system of spectral and mass units is understood in these normalization formulas. □

In particular, $\log|Z'_M| = h\log M + O(1)$ and $\log|P'_M| = (h/2)\log M + O(1)$. Removing zero modes leaves the eta phase intact, but changes the normalization needed for the modulus.

6 Extensions and scope

6.1 An abstract heat-trace criterion

The phase calculation can be separated from the odd-dimensional geometry used to establish its analytic input.

Proposition 6.1. *Let D be a self-adjoint first-order elliptic operator on a closed n-dimensional manifold. Suppose that it satisfies the Weyl estimate (2.4) and*

$$\int_0^\infty t^{-1/2} |\operatorname{Tr}(De^{-tD^2})|\,\mathrm{d}t < \infty. \tag{6.1}$$

Set

$$\eta_{\mathrm{heat}}(D) = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} \operatorname{Tr}(De^{-tD^2})\,\mathrm{d}t. \tag{6.2}$$

For any even $N > n$, the primed massless PV product satisfies

$$\lim_{M\to\infty} \operatorname{ph} Z'_M(D) = e^{-\mathrm{i}\pi\eta_{\mathrm{heat}}(D)/2}. \tag{6.3}$$

If the spectral eta function is regular at zero, then $\eta_{\mathrm{heat}}(D) = \eta_D(0)$. The negative-mass phase and the Pfaffian phases also hold with $\eta_{\mathrm{heat}}(D)$ in place of $\eta_D(0)$, with (4.1) imposed for Pfaffians.

Proof. Finite-mass absolute convergence uses only the Weyl estimate and the moment identities. The Fubini estimate (3.19) is likewise unchanged. Equation (6.1) supplies exactly the majorant required in (3.22), proving the limit. For $\Re s > 0$, the Mellin integral (2.7) is holomorphic and agrees with the spectral eta function where its defining series converges. Dominated convergence along real $s \downarrow 0$ identifies its limit with (6.2). If eta is regular at zero, this limit is its value there. The signed-mass kernel (5.4) and the Pfaffian construction prove the remaining assertions. □

The stronger estimate $T(t) = O(t^{1/2})$ directly implies regularity as in Lemma 2.2. The proposition does not extend the odd-dimensional modulus formula without additional heat-kernel information.

6.2 Oriented even-dimensional manifolds

For a nonchiral geometric Dirac operator on the full spinor bundle of a closed oriented even-dimensional manifold, the chirality operator Γ satisfies $\Gamma D = -D\Gamma$. Its nonzero spectrum is symmetric, so $T(t) = 0$ and $\eta_D(0) = 0$. Moreover, (3.10) obeys $\ell_M(-\lambda) = \overline{\ell_M(\lambda)}$. Thus the primed massless PV determinant is positive at every M, and its phase is already one before the large-mass limit.

The heat trace now has integer rather than half-integer powers. The vanishing odd-moment argument for the modulus therefore does not apply, and real local subtractions are generally needed. For negative physical mass, Proposition 6.1 gives the additional zero-mode sign in (5.2).

6.3 A covering-space criterion and Pin examples

On a nonorientable manifold, global chirality need not exist. Even-dimensional eta phases can consequently be nontrivial. The operator and its Pin-module structure must first be specified.

The following sufficient criterion is useful without making a universal assertion about every Pin Dirac convention.

Suppose $\widetilde{Y} \to Y$ is a double cover with a fixed-point-free deck involution τ. Let $\widetilde{D}$ be a lifted geometric Dirac-type operator and let U_τ be a unitary involution lifting τ, commuting with $\widetilde{D}$. Assume that D acts on the invariant sections and that a unitary grading $\widetilde{\Gamma}$ anticommutes with $\widetilde{D}$. Then

$$T_Y(t) = \frac{1}{2}\operatorname{Tr}(\widetilde{D}e^{-t\widetilde{D}^2}) + \frac{1}{2}\operatorname{Tr}(U_\tau \widetilde{D}e^{-t\widetilde{D}^2}). \tag{6.4}$$

The first trace vanishes by the grading. Compactness and the absence of fixed points give a positive lower bound on the distance from x to τx. The usual off-diagonal Gaussian estimate, also for the differentiated heat kernel, bounds the second trace by $O(t^{-L}e^{-c/t})$ for some L and $c > 0$. This proves the short-time integrability required in Proposition 6.1.

The four-dimensional Pin^+ Majorana system discussed in [1, Appendices A and C] provides the standard example of a nontrivial even-dimensional eta phase. With the operator and multiplicity conventions used there, the two Pin^+ structures on the round $\mathbb{R}P^4$ give $h = 0$ and $\eta_D(0) = \pm 1/4$. The negative-mass Pfaffian phase is accordingly $e^{\mp i\pi/8}$. This example uses the specified Pin structure and operator; parity of the dimension alone does not determine the answer.

6.4 Chiral fields, boundaries, and finite local terms

A chiral operator $D^+ : S^+ \to S^-$ is not a self-adjoint operator on a single space of sections. Its determinant, and the corresponding Pfaffian in a real theory, generally takes values in a line bundle over the background fields. Higher-dimensional eta invariants and anomaly inflow enter its global interpretation [5, 6, 2]. The scalar products constructed here cannot simply be applied to D^+ by replacing D in their formulas.

On a manifold with boundary, a self-adjoint domain and suitable heat estimates must likewise be supplied. Our main theorem is for closed manifolds. Finally, a finite imaginary local term in the effective action can change a phase. All equalities in this paper refer to the mass sign (2.1) and the explicitly specified PV normalization. Independence of N in Theorem 3.1 is an independence within that family of schemes, not a statement about arbitrary finite counterterms.

7 Discussion

The construction separates the ultraviolet cancellation from the spectral asymmetry. Finite differences make the combined rational spectral factor trace class at each regulator scale. The eta invariant then enters through an absolutely convergent heat integral, and the Bismut–Freed estimate justifies passage to infinite mass. In odd dimensions the same regulator moments annihilate the divergent heat terms in the modulus, yielding the conventional positive zeta-determinant amplitude.

For Pfaffians, the antiunitary symmetry specifies finite-dimensional antisymmetric forms on the eigenspaces. Their PV ratios fix the square root, including its spectral-flow sign. Negative physical mass changes the limiting heat kernel by a factor of two, while zero modes contribute the explicit signs in Theorem 5.1. In a massless primed determinant they instead change the power of the regulator scale required to normalize the modulus.

The finite-product convergence proofs give an explicit link between the physical eta phases of [1, 2] and the heat-kernel results of [3, 4]. Their hypotheses also make precise the distinction from the determinant-line constructions of [5, 6].

References

[1] E. Witten, *Fermion Path Integrals And Topological Phases*, Rev. Mod. Phys. **88** (2016), 035001, arXiv:1508.04715.

[2] E. Witten and K. Yonekura, *Anomaly Inflow and the η-Invariant*, arXiv:1909.08775.

[3] J.-M. Bismut and D. S. Freed, *The Analysis of Elliptic Families II. Dirac Operators, Eta Invariants, and the Holonomy Theorem*, Commun. Math. Phys. **107** (1986), 103–163. Available PDF.

[4] R. Ponge, *A new proof of the local regularity of the eta invariant of a Dirac operator*, J. Geom. Phys. **56** (2006), 1654–1665, arXiv:math/0411123.

[5] X. Dai and D. S. Freed, *Eta-Invariants and Determinant Lines*, J. Math. Phys. **35** (1994), 5155–5194, arXiv:hep-th/9405012.

[6] D. S. Freed and G. W. Moore, *Setting the Quantum Integrand of M-Theory*, Commun. Math. Phys. **263** (2006), 89–132, arXiv:hep-th/0409135.